

BLACK HAWK SOC-OS

Security Operations Platform

CLIENT	Alpha Pride Security	PREPARED BY	Immersicloud Consulting
SYSTEM	Black Hawk SOC-OS	CONTACT	immersitec@gmail.com · immersicloud.org
CATEGORY	Enterprise Security Operations Platform	SCOPE	Full-stack SaaS — Web + Mobile + AI
MARKET	Kenya / East Africa	DOCUMENT	Project Reference — Clients & Developers

1 · PLATFORM OVERVIEW

Black Hawk SOC-OS is a purpose-built **Security Operations System** designed for African private security companies operating in complex, multi-site environments. Built by Immersicloud Consulting for Alpha Pride Security Kenya, the platform unifies every operational desk — from control room dispatch to field patrols, HR, finance, and executive intelligence — into a single shared workspace.

Unlike fragmented tools (radios + spreadsheets + WhatsApp + paper logs), Black Hawk SOC-OS gives every team member — from a guard on patrol to the CEO — a role-aware, real-time view of the operation. Every action is timestamped, auditable, and feeds the intelligence layer automatically.

24 modules	<2s	99.95%	100%	4 wks
Unified in one console	Median dispatch time	Operational uptime SLA	Immutable audit chain	Deployment to go-live

What Makes It Different

- **Single shared dataset.** Modules aren't bolted on — they share state, audit trails and permissions natively.
- **Real-time everywhere.** Control room, field officers, supervisors and executives see the same live picture.
- **Offline-first field app.** Guards operate without internet; data syncs automatically on reconnect.
- **AI-powered intelligence.** Incidents are summarised, patrols are anomaly-checked, and strategic advisories are AI-scored — automatically.
- **Evidence-grade security.** Every piece of evidence is SHA-256 hashed, chain-of-custody tracked, and stored with immutable audit logs.
- **Role-aware access.** Every user sees only what their role permits — CEO to field officer, all in one login.
- **Kenya-native.** Built for Kenyan operations: M-Pesa ready, Kenya DPA compliant, NTSA/Kenya Met integrations, KES currency.
- **No integrations to manage.** CCTV, body cam, alarms, dispatch, HR, finance — one platform, zero middleware.
- **White-label client portal.** Clients get their own branded view of SLA performance, incidents, and reports.
- **4-week deployment.** Discovery → sites & hardware → training → go-live. Supported by Immersicloud every step.

2 · THE 24 MODULES

Black Hawk SOC-OS is built around six operational departments, each containing purpose-built modules. All modules share the same user database, permission engine, and audit trail — switching between them is instant and context is preserved.

Control & Dispatch

Control Room Command

10-desk command hub. Live alarm board, MDT map, SOP overlays, SLA timers, one-click dispatch. Operators get colour-tagged incident cards with full contact tree.

Alarm & Mobile Response (MDT)

Unified alarm intake + Mobile Dispatch Terminal. Haversine proximity engine routes the nearest available vehicle. Audio alerts, false-alarm tracking, GPS mapping.

Digital Occurrence Book (DOB)

Immutable, role-aware operational log replacing paper OB. Every entry is timestamped, signed, sequenced, and exportable to PDF/CSV/JSON. Soft-delete with audit.

Communications Centre

Live call handling, WhatsApp/SMS centre, dispatch escalation, radio bridge, and communications metrics. All comms linked to incidents and tickets.

Field Operations**GPS Patrol Tracking**

Real-time geofenced officer positions on live map. Heatmaps, dwell-time analysis, patrol gap detection via AI anomaly function.

Supervision Patrol

QR / NFC / RFID verified guard tours. Selfie + GPS proof of presence. Missed checkpoint alerts to supervisor in real time.

Field Officers Platform

Rank-aware mobile platform for officers. Offline-first — works without internet. Welfare check-ins, SOS button, shift briefings, MDT integration.

Escort & VIP Protection

Route risk scoring, convoy management, protective detail assignments. Real-time GPS tracking of escort missions with client visibility.

Technical Security**CCTV & Video Surveillance**

Multi-vendor camera registry (Dahua, Hikvision, ONVIF). Zone management, recording schedules, operator console, link cameras to incidents.

Body Cam Streaming & Evidence Vault

Live body cam feeds + evidence library. SHA-256 hash on every file. Chain-of-custody log. Signed URL downloads with access audit. Phase 6 hardening applied.

Access Control

Visitor pre-clearance, badge management, watchlist screening, entry/exit log. Armoury custody chain for firearms, radios, and equipment.

Technical Security / TIMU

Job cards, device health monitoring, planned maintenance schedules, SLA tracking, performance analytics, quality inspections.

Intelligence & Advisory**Incident Command Centre**

Full incident lifecycle: open → investigating → resolved → closed. SLA countdown ring, escalation tree (Supervisor→Ops Mgr→COO→CEO), evidence attachment, AI auto-summary, SOP checklist, timeline, PDF export.

Investigations

Case files with full evidence chain-of-custody. Witness statements, lab submissions, suspect tracking, investigation timeline, close-rate analytics.

Strategic Advisory (RTSI)

Real-time Strategic Intelligence: Traffic, Protest, Terror, Weather, Crime advisories. Mapbox globe with NORMAL/CAUTION/CRITICAL colour coding. Ingests NTSA, Kenya Met, DCI, OSINT feeds. AI-scored threat analysis.

Analytics & Executive Reports

KPI scorecards, loss-control intelligence, patrol compliance trends, SLA performance charts, board-ready PDF exports. Role-specific dashboards per designation.

Specialised Departments**Cash In Transit (CIT)**

Full CIT lifecycle: runs, manifests, vault movements, route risk scoring, crew management, in-transit incidents. Declared value tracking in KES.

K9 Unit Management

Dog registry (breed, certifications, welfare), handler assignments, deployment log, K9 incident reporting, health tracking.

Event Security

Event creation with risk assessment, staff assignment, task management, real-time monitoring dashboard, emergency panel, incident reporting, client communications, post-event reports.

Courier Operations

Chain-of-custody dispatch, rider/driver management, delivery tracking, route management, incident reporting, client delivery confirmation.

Management & HR	
Staff & HR Suite Full HR lifecycle: recruitment pipeline, onboarding checklists, attendance tracking, performance reviews (1–5 scale), disciplinary records, leave management.	Payroll & Finance Payroll runs, payslip generation, PAYE/NHIF/NSSF/NITA statutory calculations. Billing & invoicing with recurring contracts, bank reconciliation, expense claims, profit & loss.
Training Management Certification matrix, training programme scheduling, e-learning module tracking, drill simulator (4 seeded scenarios: perimeter breach, panic, armed robbery, medical).	Client Management Client sites, risk tier classification, service contracts, client portal with branded SLA scoreboard. Billing rules, deposits, statements, purchase orders.

3 · AI INTELLIGENCE LAYER

Black Hawk SOC-OS embeds AI into the operational workflow — not as a bolt-on chatbot, but as a silent analyst running continuously in the background. Every AI function is built as a Supabase Edge Function (Deno TypeScript) calling OpenAI GPT-4o.

AI FUNCTION	TRIGGER	OUTPUT / VALUE
Incident AI Summary	Manual button on any incident	AI-generated incident brief with timeline, evidence links, and SLA performance. Saves 30–45 min per major incident.
Patrol Anomaly Detection	Runs automatically on patrol data	Flags missed checkpoints, unusual dwell times, route deviations. Alerts supervisor before the shift ends.
Strategic Advisory (AI Scoring)	Live feed refresh (NTSA, Kenya Met, DCI, OSINT)	Each advisory scored NORMAL / CAUTION / CRITICAL with recommended action. Feeds the RTSI map and alert engine.
Shift Briefing Generator	Start of each shift	Auto-generated briefing covering overnight incidents, open alarms, patrol compliance, priority sites, weather advisories.
Threat Analysis	Linked to strategic advisories	Classifies threat type, geographic scope, and recommended escalation level. Feeds the War Room and COO dashboard.
Risk Forecast	Weekly automated run	Predicts risk scoring per client site based on historical incidents, patrol compliance, and external threat data.
Attendance Anomaly Detection	Daily payroll / scheduling check	Flags ghost workers, unusual absence patterns, and AWOL cases before payroll runs.
AI Copilot (SOC Assistant)	On-demand — any operator can ask	Natural language Q&A; over live operational data. 'Which sites have open incidents older than 48h?' 'Who is on duty at JKIA tonight?'
Training Drill Injector	Supervisor-initiated drill	Injects a realistic drill scenario (perimeter breach, panic alarm, armed robbery, medical) into the live feed for tabletop simulation.

4 · ROLES & ACCESS CONTROL

The platform enforces a dual-layer security model: **frontend permission gating** (every button, tab, and route checks the user's role before rendering) and **database Row Level Security** (Postgres RLS policies mirror the frontend matrix, so even a direct API call cannot bypass access rules). Roles are assigned in the database — never client-side.

Permission Levels

LEVEL	CAPABILITY	TYPICAL HOLDERS
none	Route blocked, element hidden, API rejected	Any user outside the module's permitted roles
view	Read-only — see data, cannot modify	CEO overview of HR; client portal users
edit	Read + update existing records	Control room operators, field officers
create	Read + update + create new (implies edit)	Supervisors, HR officers, finance officers
delete	Full CRUD including irreversible operations	Executives (CEO, COO), system admin

Role Directory

ROLE	ACCESS SUMMARY	ROLE	ACCESS SUMMARY
CEO	Executive dashboard (full CRUD), all other modules view-only	Payroll Officer	Pay runs, payslips, PAYE/NHIF/NSSF/NITA statutory
COO	Full operational control: control room, incidents, alarms, fleet, CIT, patrol, MDT	CIT Manager / Officer	CIT runs, manifests, vault, routes, crews, incidents
General Manager	Full client management + contracts. Finance and ops view.	Courier Manager	Full courier ops: deliveries, dispatch, riders, routes
Operations Manager	Full ops: dispatch, incidents, alarms, MDT, patrol, scheduling	Branch Manager	Branch-scoped view of ops, clients, staff, billing
Control Room Officer	Control room (full), incidents (create), alarms (edit), CCTV/body cam (view)	Compliance Officer	Full governance: compliance register, audit log, policies, approvals
HR / HR Officer	Full HR: staff records, leave, training, documents, scheduling	System Admin	Users, roles, system settings, tenants — NO operational data access
Finance / Finance Officer	Full finance: billing, invoicing, bank rec, expenses, loss control	CS Manager / Officer	Customer service tickets, complaints, team management

5 · SECURITY & COMPLIANCE

STANDARD / FEATURE	IMPLEMENTATION	STANDARD / FEATURE	IMPLEMENTATION
Encryption at rest	AES-256 — all data encrypted at rest in Supabase-managed Postgres	Kenya Data Protection Act	Data stored in tenant-isolated schemas. Client retains full data ownership
Encryption in transit	TLS 1.3 on all client-server connections, including edge functions	SSO / Biometric ready	Supabase Auth supports OAuth 2.0 SSO; biometric hook points built into field app
Access control	RBAC + Postgres Row Level Security — database enforces permissions even on direct API calls	Multi-tenant isolation	Each organisation's data is logically isolated. No cross-tenant data leakage possible
Evidence integrity	SHA-256 hash stored on every evidence file. Any tamper is immediately detectable	Session security	Global signout revokes all sessions server-side. Tokens are short-lived and auto-refreshed
Audit trail	Immutable audit log on all create/update/delete operations. Cannot be altered once written	On-premise option	Full on-premise or hybrid deployment available for banks, government, sovereign clients

6 · DEPLOYMENT MODELS

MODEL	IDEAL FOR	LATENCY	KEY BENEFITS
■ Cloud SOC (Multi-tenant SaaS)	Mid-market & residential estate portfolios	80–120ms	Zero on-site infrastructure · Auto-scaling · Shared 24/7 monitoring · Fastest deployment
■ On-Premise (Client-hosted)	Banks, government, embassies, air-gap environments	Sub-10ms (LAN)	Full data sovereignty · Air-gap option · Dedicated SOC desk · Maximum security
■ Hybrid Edge (On-site + cloud HQ)	National operators, multi-site enterprises, remote sites	Edge real-time, cloud sync	Edge survives outages · Centralised intelligence · Local speed + cloud analytics

4-Week Deployment Timeline

WEEK	PHASE	ACTIVITIES	DELIVERABLE
Week 1	Discovery	Stakeholder workshops · current-state audit · module scoping · access matrix design · data migration assessment	Signed scope document + access matrix
Week 2	Sites & Hardware	QR/Rfid checkpoint install · body-cam provisioning · CCTV integration mapping · staff database import · client site seeding	Live test environment with real data
Week 3	Training	Control room operator drills (2–3 days to proficiency) · supervisor enablement · field officer MDT certification (90 min) · client portal walk-through	Certified operator cohort

WEEK	PHASE	ACTIVITIES	DELIVERABLE
Week 4	Go-Live	Cutover from legacy systems · parallel run · 24/7 Immersicloud white-glove support · incident response drill · handover to Alpha Pride ops team	Full production cutover

7 · OPERATIONAL OUTCOMES

The following outcomes are based on the platform's built-in automation, response workflows, and intelligence layer — derived from the operational capabilities of Black Hawk SOC-OS.

OPERATIONAL AREA	BEFORE	WITH BLACK HAWK SOC-OS	MECHANISM
Response time	Manual radio dispatch, 9+ minutes average	Median <2 minutes	Proximity engine + auto-dispatch + MDT push notification
False alarm rate	High — no verification workflow	61% reduction (target)	Structured resolve workflow with false-alarm flag + pattern tracking
Patrol compliance	Paper sign-off, easily falsified	100% verifiable	QR/NFC scan + GPS selfie + timestamped DOB entry
Incident report time	Hours to draft manually	Auto-generated in <30s	AI incident summary from structured data + evidence links
Investigation close rate	Low — fragmented evidence	3× faster (target)	Structured case file, evidence vault, chain-of-custody, timeline
Client satisfaction	Reactive, no visibility	+38 points NPS (target)	Branded client portal with live SLA scoreboard and incident feed
CIT losses	Manual manifests, no real-time tracking	Zero in-transit losses (pilot target)	Real-time GPS + sealed manifest + vault custody chain
Evidence integrity	Photos on WhatsApp, no chain of custody	100% tamper-evident	SHA-256 hash + signed URL access + immutable access log
Audit readiness	Days to produce audit trail	Instant export	Immutable audit log on every action, exportable to PDF/CSV

8 · TECHNICAL ARCHITECTURE [DEVELOPER REFERENCE]

This section is intended for developers, technical leads, and IT teams. It describes the exact technology choices, architecture patterns, and infrastructure that power Black Hawk SOC-OS.

Stack Overview

LAYER	TECHNOLOGY	PURPOSE
Frontend Framework	React 18 + TypeScript + Vite 5	UI layer — all pages, components, forms, real-time views
Styling	Tailwind CSS + shadcn/ui (Radix)	Utility-first styling with accessible Radix component primitives
Routing	React Router v6	Client-side routing with nested layout + role-gated routes
Server State	TanStack Query v5	All DB reads via React Query — caching, background refresh, optimistic updates
Database	Supabase · Postgres 15	75+ tables, JSONB columns, full-text search, pg_cron, Row Level Security
Auth	Supabase Auth (JWT + RLS)	Email/password + SSO-ready. Server-side roles in user_roles table.
Edge Functions	Deno TypeScript (Supabase Functions)	14 AI-powered functions — incident summary, patrol anomaly, advisory, briefing generation
AI / LLM	OpenAI GPT-4o via Edge Functions	Structured JSON prompts + function calling for reliable operational outputs

LAYER	TECHNOLOGY	PURPOSE
Realtime	Supabase Realtime (WebSocket)	Postgres change events push to all subscribed clients — no polling
File Storage	Supabase Storage (S3-compatible)	Evidence vault, staff photos, body cam clips, report PDFs
Maps	Mapbox GL	GPS patrol tracking, RTSI globe, escort route mapping, CCTV zone overlays
PDF Generation	jsPDF + jspdf-autotable	In-browser PDF export for incidents, payslips, investigation reports, client briefs
Forms	React Hook Form + Zod	Schema-validated forms across all modules
Animation	Framer Motion	Page transitions, stat card animations, alert pulse effects
Offline Sync	Custom useOfflineSync hook	Field officer actions queued locally and synced on reconnect

Architecture Patterns

Data Flow

Auth & Security

Page Architecture

Database Conventions

- All DB reads go through **TanStack Query** hooks — never directly in component body
- Each module has a dedicated **use[Module].ts** hook owning all data + mutations
- Realtime subscriptions open once in the hook via **supabase.channel()** and tear down on unmount
- Mutations call **queryClient.invalidateQueries()** on success — no manual state updates
- All pages lazy-loaded via **safeLazy()** — handles chunk errors with auto-reload
- Every page must have a **default export** — **safeLazy()** requires it
- Page component owns only **local UI state** (dialogs, filters) — data comes from hook
- Shared shell: **WorkspaceShell** renders sidebar + topbar; never re-mount on navigation
- Auth state set ONLY from **onAuthStateChange** — never from getSession() (prevents race conditions)
- All PKs: **UUID PRIMARY KEY DEFAULT gen_random_uuid()**
- DB role fetched server-side on login — never read from sessionStorage or local state
- All timestamps: **TIMESTAMPTZ DEFAULT NOW()**
- **RequireRole** wraps routes; **RequirePermission** wraps individual UI elements
- Auth FK always to **public.profiles** — never directly to auth.users
- RLS policies mirror the frontend access matrix — API calls cannot bypass permissions
- Every new table: **ALTER TABLE ... ENABLE ROW LEVEL SECURITY**
- Migrations always idempotent: **CREATE TABLE IF NOT EXISTS**

9 · BUILD STATUS & DELIVERY ROADMAP [DEVELOPER REFERENCE]

Current state: **frontend is feature-complete** across all 24 modules. Backend (Supabase schema, RLS policies, edge functions) is partially applied. The delivery roadmap below shows the remaining backend work, ordered by operational priority.

✓ COMPLETE	~ IN PROGRESS	! PENDING	Frontend complete + real DB data wired · Frontend done, DB partially wired · DB tables not yet created
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PHASE	REF	MODULE	STATUS	CURRENT STATE	NEXT ACTION
Phase A · Foundation	A1	User Management & Roles	✓ COMPLETE	Auth flow complete. user_roles table and app_role enum applied. RequireRole + RequirePermission guards active.	Add admin UI to assign roles and build PendingActivation screen for users awaiting role assignment.
Phase A · Foundation	A2	Sites, Clients & Staff Seed	! PENDING	Schema exists (sites, clients, staff tables applied). Seed SQL files are empty.	Write seed/001_sites.sql with real Alpha Pride client sites. Write seed/002_staff.sql with 20+ staff records. Run seeds.
Phase A · Foundation	A3	Staff Management (full CRUD)	~ IN PROGRESS	StaffManagement page (353 lines) partially reads real DB. Photo upload not yet wired to storage.	Build useStaff.ts hook (list, create, update, deactivate). Wire photo upload to Supabase Storage bucket staff-photos.
Phase B · Daily Ops	B1	Alarms & Auto-Dispatch	~ IN PROGRESS	Alarms page (428 lines) + useAlarms hook fully built. Phase 7 migration (auto_dispatch_rules) not yet applied.	Apply Phase 7 migration SQL. Wire auto-dispatch rule engine. Wire false_alarm_log. Test full alarm lifecycle end-to-end.
Phase B · Daily Ops	B2	Digital Occurrence Book	~ IN PROGRESS	DOB page (340 lines) + useDOBEentries hook built and reads real DB. Soft-delete migration pending.	Write and apply dob_soft_delete.sql (add deleted_at, deleted_by columns). Configure DOB auto-entry from incident/alarm triggers.
Phase B · Daily Ops	B3	Patrol Suite (full backend)	~ IN PROGRESS	Shell tab wrapper built (72 lines). GPS tracking, supervision patrol, and patrol intelligence sub-pages exist as UI stubs.	Write field_portal_phase2.sql (patrol_sessions, checkpoint_scans, patrol_gaps view). Build usePatrols.ts. Wire anomaly detection edge function.
Phase B · Daily Ops	B4	Communications Centre	~ IN PROGRESS	73-line shell + 8 sub-components (~200 lines each) built. All components currently use mock/static data.	Build useComms.ts covering calls, tickets, whatsapp_messages, sms_messages tables. Wire all 8 sub-components to real data + realtime.
Phase B · Daily Ops	B5	Control Room & Dispatch	~ IN PROGRESS	ControlRoom page (328 lines) built. Dispatch not fully wired. hq_connect.sql (directives) not applied.	Write hq_connect.sql (directives, directive_acknowledgements). Build useDispatch.ts. Wire MDT to mdt_messages table.
Phase C · Specialist	C1	CCTV Camera Registry	! PENDING	CCTVOperatorConsole component (434 lines) built. No cctv_cameras table exists in DB.	Write and apply cctv_cameras.sql (camera_code, label, site_id, zone, stream_url, status, retention). Wire operator console.
Phase C · Specialist	C2	Body Cam & Evidence Vault	~ IN PROGRESS	58-line shell + 4 components built. Phase 6 migration (evidence vault hardening, SHA-256) written but not applied.	Apply Phase 6 SQL from PHASE6_MIGRATION.md. Create evidence-vault storage bucket. Wire EvidenceLibrary to real DB. Deploy evidence-sign-url edge function.

PHASE	REF	MODULE	STATUS	CURRENT STATE	NEXT ACTION
Phase C · Specialist	C3	Armoury & Visitor Access	! PENDING	ArmouryCustodyModule (269 lines) and VisitorAccessModule (219 lines) built. No DB tables exist.	Write and apply visitor_access_armoury.sql (visitor_pre_clearances, visitor_log, armoury_items, armoury_custody_log). Wire both modules.
Phase C · Specialist	C4	K9 Module	! PENDING	K9 page (367 lines of UI) complete. k9_module.sql not written.	Write and apply k9_module.sql (k9_dogs, k9_handlers, k9_deployments, k9_incidents). Build useK9.ts. Wire K9 page.
Phase C · Specialist	C5	Cash In Transit (CIT)	! PENDING	CashInTransit page (632 lines) complete with all 6 tabs. cit_module.sql not written.	Write and apply cit_module.sql (cit_runs, cit_manifests, cit_vault_movements, cit_routes). crew_members table already exists. Wire all 6 tabs.
Phase C · Specialist	C6	War Room / CMC	! PENDING	WarRoom page (198 lines) + CMC components built. cmc_module.sql not written.	Write and apply cmc_module.sql (cmc_activations, cmc_decisions, cmc_resources). Wire War Room and CMC Activate Dialog.
Phase D · HR & Finance	D1	Full HR Suite	! PENDING	All 5 HR sub-pages (recruitment, onboarding, attendance, performance, disciplinary) are UI-only. Payroll tables missing.	Apply hr_finance_payroll_governance.sql (payroll_runs, payslips, expense_claims, disciplinary_records, performance_reviews). Build useHR.ts. Wire all pages.
Phase E · Intelligence	E1	Deploy Edge Functions	! PENDING	All 14 edge functions written and code-complete. OPENAI_API_KEY not set. Functions not deployed.	Set OPENAI_API_KEY in Supabase project secrets. Run: supabase functions deploy for all 14 functions. Wire Copilot drawer and Strategic Advisory UI.
Phase E · Intelligence	E2	RTSI Strategic Map	~ IN PROGRESS	StrategicAdvisory page (311 lines) + 8 sub-components built. strategic_advisories table exists. Edge functions not deployed.	Deploy strategic-advisory, threat-analysis, fetch-advisories functions. Configure Mapbox token. Test NORMAL/CAUTION/CRITICAL scoring end-to-end.
Phase F · Governance	F	Compliance, Audit, Approvals	! PENDING	Compliance (83 lines), ComplianceRegister (113 lines), ApprovalsInbox (79 lines) are UI stubs. Phase 7 audit_log not applied.	Apply Phase 7 SQL (audit_log, user_presence, auto_dispatch_rules). Wire Compliance Register, Approvals Inbox, Directive Log, SOP Library.

10 · PENDING DATABASE MIGRATIONS [DEVELOPER REFERENCE]

The following SQL migration files exist in `.lovable/pending_migrations/` but are currently empty — they need to be written and applied via **Lovable Cloud** → **Database** → **SQL Editor**. Key rules: always use `CREATE TABLE IF NOT EXISTS`, always `ENABLE ROW LEVEL SECURITY` on every new table, always `DROP POLICY IF EXISTS` before creating.

FILE	TABLES TO CREATE	PHASE	BLOCKS
cit_module.sql	cit_runs, cit_manifests, cit_vault_movements, cit_routes	C5	CIT module
k9_module.sql	k9_dogs, k9_handlers, k9_deployments, k9_incidents	C4	K9 module
visitor_access_armoury.sql	visitor_pre_clearances, visitor_log, armoury_items, armoury_custody_log	C3	Access Control, Armoury
cmc_module.sql	cmc_activations, cmc_decisions, cmc_resources	C6	War Room / CMC
cctv_cameras.sql	cctv_cameras (code, label, site_id, zone, stream_url, status, retention)	C1	CCTV module
field_portal_phase2.sql	patrol_sessions, checkpoint_scans, patrol_gaps (view)	B3	Patrol Suite
hq_connect.sql	directives, directive_acknowledgements	B5	Control Room, Dispatch
hr_finance_payroll_governance.sql	payroll_runs, payslips, expense_claims, disciplinary_records, performance_reviews	D1	HR Suite, Payroll
auto_dispatch_rules.sql	auto_dispatch_rules (Part of Phase 7 migration)	B1	Alarm auto-dispatch
dob_soft_delete.sql	ALTER TABLE dob_entries ADD deleted_at, deleted_by	B2	DOB module
incident_command_v2.sql	incident_command_structure, ics_task_assignments	C	Incident Command upgrade
enterprise_first_wave.sql	Multi-branch entity structures, branch registry	H	Multi-tenancy / scale-out
directives_audit.sql	directive_log, acknowledgement_records	F	Governance / Compliance

11 · QUICK REFERENCE [DEVELOPER REFERENCE]

TASK	FILE / COMMAND	TASK	FILE / COMMAND
Add a new route	src/App.tsx → ConsoleLayout routes	Call edge function	supabase.functions.invoke('name', { body })
Add a sidebar link	src/components/shell/Workspaces/hell.tsx	See which tables exist	src/integrations/supabase/types.ts
Add a permission	src/config/accessControl.ts → ModuleKey + matrix	Apply a migration	Lovable Cloud → Database → SQL Editor
Guard a route	in App.tsx	Deploy edge function	supabase functions deploy
Guard a button/tab		Set AI API key	Supabase Dashboard → Settings → Secrets → OPENAI_API_KEY
Write to audit trail	import { logAudit } from '@utils/auditLog'	Lazy load a page	safeLazy('Name', () => import('./pages/Name'))
Show a toast	import { toast } from 'sonner'	Check user role	useAuth().userRole – never sessionStorage
Export PDF in browser	import jsPDF from 'jspdf' + autoTable	Check permission	usePermissions().can('module.key', 'edit')

